

A Comparative Study of N-Body Simulation and Trajectories in the Milky Way & Andromeda Collision

Field of study: Physics

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Abstract

The objectives of this research article are to develop a mathematical model for simulating the collision phenomenon between the Milky Way and Andromeda galaxies with high realism using limited computational resources, and to study dynamics, trajectories and verify the collision trend of both galaxies in the future. The research methodology involves creating an N-body simulation using Python, applying the Leapfrog Integration method combined with Gravitational Softening technique, and optimizing processing with Parallel Computing to simulate the system of 5,801 particles. The resulting coordinate data was then used to generate 3D animation using Blender software, and the model's validity was verified based on the law of conservation of energy and angular momentum. The research results found that developers successfully created an efficient N-body model with a total energy drift error of only 6.12 percent, which is within an acceptable range. The model demonstrated dynamic behaviors consistent with astrophysical theories. Furthermore, the collision patterns of the galaxies depend on initial parameters, specifically the impact angle and transverse velocity. If the impact angle is less than 30 degrees, there is a high probability of a complete merger, provided that the model can simulate results consistently with both the original hypothesis from Hubble Space Telescope data (a head-on collision) and new data from the GAIA satellite (a glancing fly-by).

Keywords: Milky Way Galaxy, Andromeda Galaxy, N-body simulation, Python, Galaxy collision simulation

I. INTRODUCTION

The Milky Way and Andromeda are the two most massive galaxies in the Local Group. Based on observational data from the Hubble Space Telescope, these two galaxies are approaching each other at a relative velocity of approximately 110 km/s, with a predicted collision and subsequent merger in about 4 billion years (Sohn et al., 2012). However, the structural transformation involved in this galactic merger spans billions of years, far exceeding the human lifespan and rendering direct observation of the complete phenomenon impossible. Furthermore, the study of galactic dynamics is governed by the N-body problem which poses immense mathematical

complexity and lacks an exact analytical solution (Binney & Tremaine, 2008).

Consequently, computer simulations utilizing numerical methods have become indispensable tools for astrophysicists to comprehend cosmic evolution, allowing for the iterative computation of positions and velocities of individual particles under mutual gravitational influence.

To address this, this research project aims to develop a mathematical model that simulates the trajectories of the collision between the Milky Way and Andromeda galaxies. The model is designed to achieve the highest possible degree of realism within the constraints of available computational resources. A primary objective is to investigate how varying initial conditions influence the resulting collision patterns of the two galaxies. To enhance comprehensibility and provide a holistic view of the macroscopic phenomenon, the simulation outputs are rendered into three-dimensional animations. Finally, to ensure the validity and accuracy of the model, the simulation

results are strictly cross-verified against fundamental physical conservation laws, as well as modern observational data from the Hubble and Gaia space telescopes.

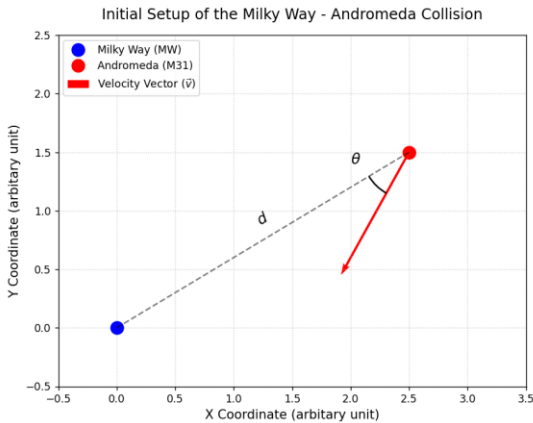
II. METHODOLOGY

To develop a mathematical model for studying the dynamics and orbital trajectories in the collision between the Milky Way and Andromeda galaxies, this study was conducted in four primary phases: data acquisition, computational modeling, system validation, and 3D visualization.

A. Data Acquisition and Initial Conditions

Primary observational data, including radial velocity, transverse velocity, and rotational speeds of the Andromeda galaxy, were gathered from the Hubble Space Telescope and Gaia satellite missions. Secondary physical parameters of both galaxies were sourced from existing astrophysical literature. The collected data were subsequently filtered and formatted for computational input.

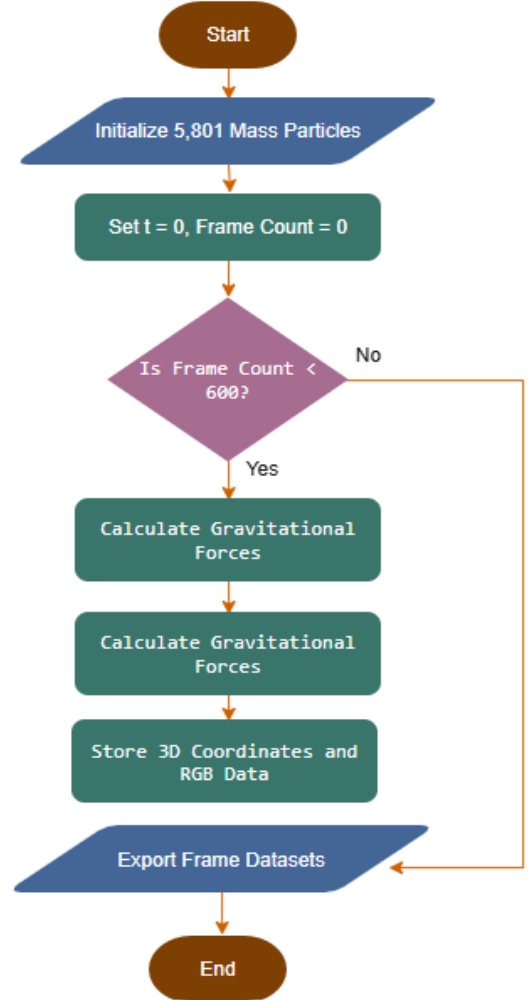
Initial Setup Diagram



B. N-Body Simulation and Computational Techniques

The computational model was developed using Python to generate instantaneous spatial frames. The galactic systems were

modeled using a total of 5,801 mass particles, where each particle represents a massive cluster of stellar material and dark matter rather than an individual star. Each frame dataset contains 3D Cartesian coordinates and RGB color codes to distinguish different galactic regions. To produce a 10-second animation at 60 frames per second (fps), 600 instantaneous frames were computed.



To ensure computational stability and prevent infinite gravitational forces during close particle encounters (singularities), a gravitational softening parameter was imposed into the force calculation.

$$\mathbf{F}_i = Gm_i \sum_{j \neq i} \frac{m_j(\mathbf{r}_j - \mathbf{r}_i)}{(|\mathbf{r}_j - \mathbf{r}_i|^2 + \epsilon^2)^{\frac{3}{2}}}$$

where ϵ is gravitational softening

Furthermore, Leapfrog integration was implemented to accurately compute particle trajectories. This symplectic integrator was chosen for its time-reversibility, which is crucial for conserving the system's total energy over long-term integration.

```

144 def update(frame): 1usage
145     global pos, vel, acc
146
147     # Physics Update (Leapfrog)
148     vel += acc * dt / 2.0
149     pos += vel * dt
150     acc = compute_accel_numba(pos, mass, G, eps_sq)
151     vel += acc * dt / 2.0
152
153     # Update Graphic
154     scatter.set_offsets(pos[:, :2])
155     time_text.set_text(f'Simulation Time: {frame * dt:.2f}')

```

To optimize CPU processing time and achieve performance comparable to C-compiled languages, parallel computing was executed using the Numba library via a Just-In-Time (JIT) compiler.

C. System Validation and Conservation Laws

A secondary analytical script in Python was developed to validate the physical accuracy of the generated frames. Data from both solitary spiral galaxy simulations and the colliding pair were evaluated against three fundamental criteria: conservation of total energy, conservation of total angular momentum, and adherence to Keplerian trajectories during the pre-collision phase.

Energy Conservative Law

$$E_{total} = \sum_i \frac{1}{2} m_i v_i^2 + \sum_{i \neq j} \frac{G m_i m_j}{\sqrt{r_{ij}^2 + \epsilon^2}}$$

Angular Momentum Conservative Law

$$L_{total} = \sum_{i=1}^N L_i = \sum_{i=1}^N m_i (r_i \times v_i)$$

Meeting these criteria confirmed the physical accuracy and numerical stability of the underlying gravitational algorithm before proceeding to the final visualization phase.

D. 3D Visualization and Morphological Analysis

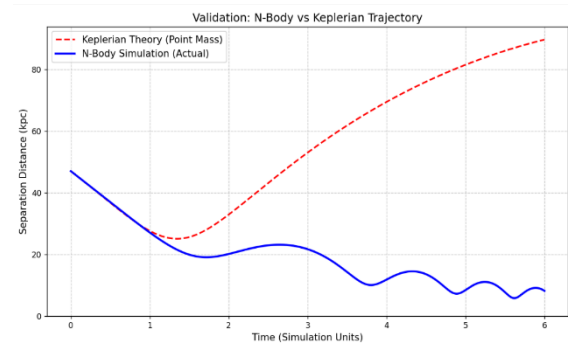
In the final phase, the verified frame datasets were imported into Blender (version 4.4) to render a 3D physical animation. This visual model illustrates the morphological evolution of the galaxies during the collision, such as the formation of tidal tails and the eventual galactic core merger. The resulting visualizations were then comparatively analyzed against theoretical frameworks and prior research findings.

III. RESULTS AND DISCUSSION

Through data analysis, model validation, and the physical simulation of the collision between the Milky Way and Andromeda galaxies, the results were compared against theoretical frameworks and existing literature. The findings are presented as follows.

A. Model Validation via Center of Mass Distance

The physical model of the galactic collision was validated by comparing the distance between the centers of mass of the two galaxies in the simulation with theoretical predictions. The results are as follows:



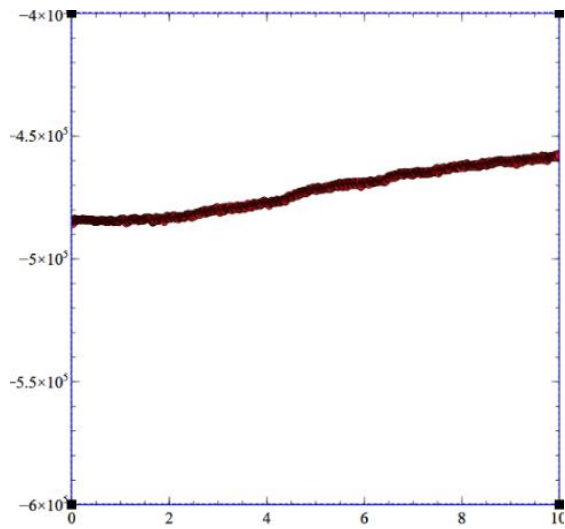
Graph 1: Distance between the centers of mass of the Milky Way and Andromeda galaxies from the physical simulation (blue line) and Keplerian mechanics theory (red dashed line).

As shown in Graph 1, the theoretical predictions and simulation results align closely from the beginning of the simulation up to

the time interval of 1 unit, with the two lines perfectly overlapping. During this pre-collision phase, the distance can be accurately calculated using Keplerian mechanics by approximating each galaxy as a point mass. However, beyond this period, the simulation graph exhibits an oscillating pattern, gradually approaching zero over time, whereas the theoretical graph diverges from zero. This divergence indicates that post-collision, all particles interact with one another, forming a gravitationally bound system. Consequently, the centers of both galaxies gradually merge into a single center of mass, rendering the theoretical point-mass approximation invalid.

B. Energy and Angular Momentum Conservation

The validity of each simulation frame was further verified using the law of conservation of total energy for a solitary spiral galaxy system via a Python validation script.

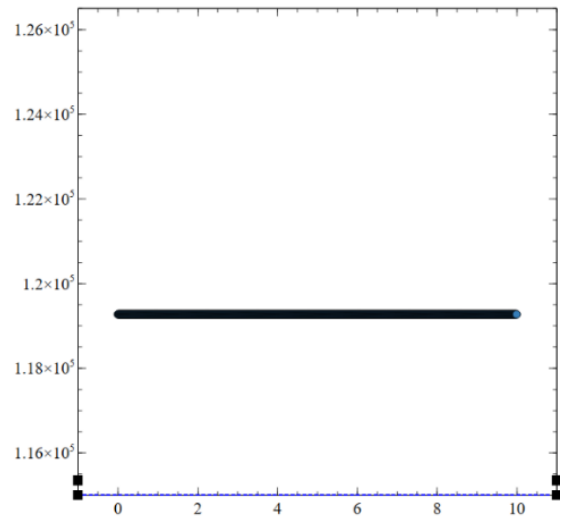


Graph 2: Total energy of a solitary spiral galaxy system.

Graph 2 reveals a gradual increase in the total energy of the spiral galaxy system over time, despite the expectation of constancy under the law of conservation of energy. The calculated energy drift is approximately 6.12%, meaning the total energy at the end of the simulation deviates from the

initial value by 6.12%. This energy fluctuation arises from the limitations of numerical integration in regions with high particle density. Nevertheless, this margin of error remains within an acceptable threshold, indicating that the model retains a sufficient degree of accuracy. In physical reality, a solitary galaxy system does not strictly adhere to perfect energy conservation, as minor energy dissipation naturally occurs. However, the assumption of overall energy conservation remains a valid approximation.

Furthermore, the conservation of total angular momentum for the solitary spiral galaxy system was verified.



Graph 3: Total angular momentum of a solitary spiral galaxy system.

As depicted in Graph 3, the total angular momentum of the solitary galaxy system remains constant over time, adhering to the law of conservation of angular momentum. This confirms that the system is free from external torques, thereby validating the high accuracy and stability of the simulation model.

C. Morphological Evolution and Simulation Timesteps

The simulated collision between the Milky Way and Andromeda was rendered into a

3D animation. The visual results at various time intervals are compiled below.

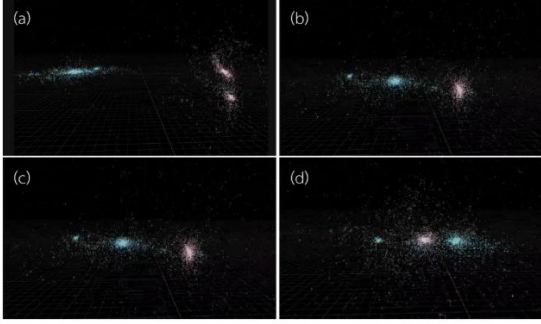


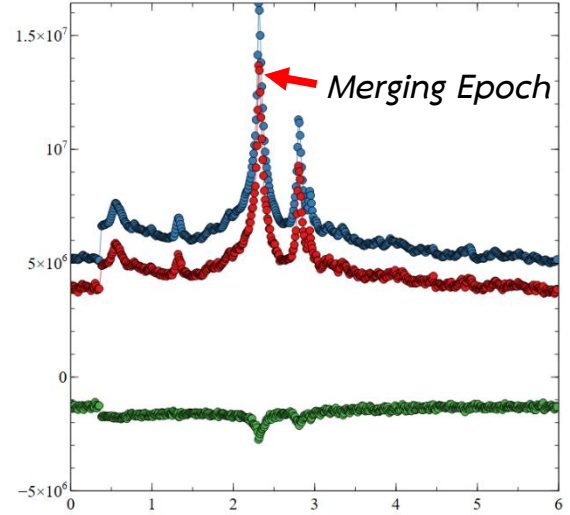
Figure 1: Physical simulation at various timesteps: (a) Initial state, (b) After 2.0 billion years, (c) After 3.5 billion years, and (d) After 5.0 billion years.

The simulation demonstrates that the morphological evolution of the galactic collision occurs gradually and can be categorized into four distinct phases:

1. **Approach Phase:** The two galaxies move toward each other following Keplerian trajectories under mutual gravitational influence.
2. **First Pericenter Passage:** Occurring at approximately 2.3 billion years, tidal forces strip material from the original structures, forming elongated tidal tails.
3. **Energy Dissipation Phase:** The distance between the two galaxies decreases rapidly due to kinetic energy dissipation and linear momentum transfer.
4. **Complete Merger:** At approximately 5.5 billion years, the galactic cores coalesce into a single center, resulting in the formation of a newly established elliptical galaxy.

D. Energy Dynamics During the Collision

An analysis of the total energy of the galactic system from the start to the end of the simulation yielded the following energy profile.



Graph 4: Energy variations of the system, including kinetic energy (blue dots), gravitational potential energy (green dots), and total energy (red dots).

According to Graph 4, all energy types remain relatively constant during the pre-collision phase. However, at approximately 2.3 billion years, the kinetic energy of the system spikes dramatically, while the gravitational potential energy experiences a significant drop. This occurs because the gravitational softening technique cannot fully counteract the intense gravitational fields in high-density regions. Consequently, localized particle collisions and high-velocity ejections occur, manifesting as sharp spikes in the energy graph.

E. Influence of Initial Conditions on Merger Morphology

The influence of initial conditions, specifically the impact angle, on the collision patterns is detailed in Table 1.

Experimental Case	Impact Angle	Time First Pericenter Passage	Time of Complete Merger	Remarks
Prograde Merger (Low Angle)	$<30^\circ$	2.15 Gyr	4.20 Gyr	Fast
Polar Merger (High angle)	$>30^\circ$	2.25 Gyr	6.85 Gyr	Delayed

Table 1: Collision patterns resulting from varying impact angles.

IV. CONCLUSION AND RECOMMENDATIONS

A. Conclusion The developed physical model effectively simulated the collision between the Milky Way and Andromeda galaxies. It accurately replicated Keplerian orbital trajectories during the pre-collision phase and realistically depicted the complex dynamics following the first pericenter passage using an N-body simulation framework. Notably, this high degree of realism was achieved within the constraints of standard personal computing resources.

The validity of the model was rigorously verified prior to the final rendering by analyzing individual frame data, and post-simulation by comparing the outcomes with existing literature. Fundamental physical conservation laws were employed as the primary validation criteria, confirming the model's accuracy and stability. Furthermore, the simulation demonstrated that the ultimate coalescence of these two spiral galaxies results in the formation of an elliptical galaxy. The study also established a direct relationship between the initial impact angle and the merger timescale; specifically, a larger impact angle significantly prolongs the time required for a complete merger.

B. Recommendations for Future Work Despite its successes, the current model possesses certain limitations that present opportunities for future enhancement. Future research should conduct a higher-resolution analysis of a broader range of impact angles. Additionally, incorporating gas dynamics (hydrodynamics) into the computational framework would allow the model to more accurately predict the luminosity and color evolution of the interacting galaxies during the merger process. Finally, migrating the core simulation codebase from Python to a lower-level compiled language, such as C or C++, is highly recommended

to accelerate computational speed and maximize the overall performance of the simulation.

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